Contact Dynamics

Qualify one passive contact without hiding its raw evidence

Observe, qualify, suppress repeats, release

Lesson	037 — component	**Time**	120–180 minutes
Prerequisites	Lessons 002, 006, and 007	**Board**	Arduino Mega 2560
Evidence	TP-C and three LED witnesses	**Status**	host verified; bench open
Energy class	E1 passive contact fixture; E0 host trace separate		

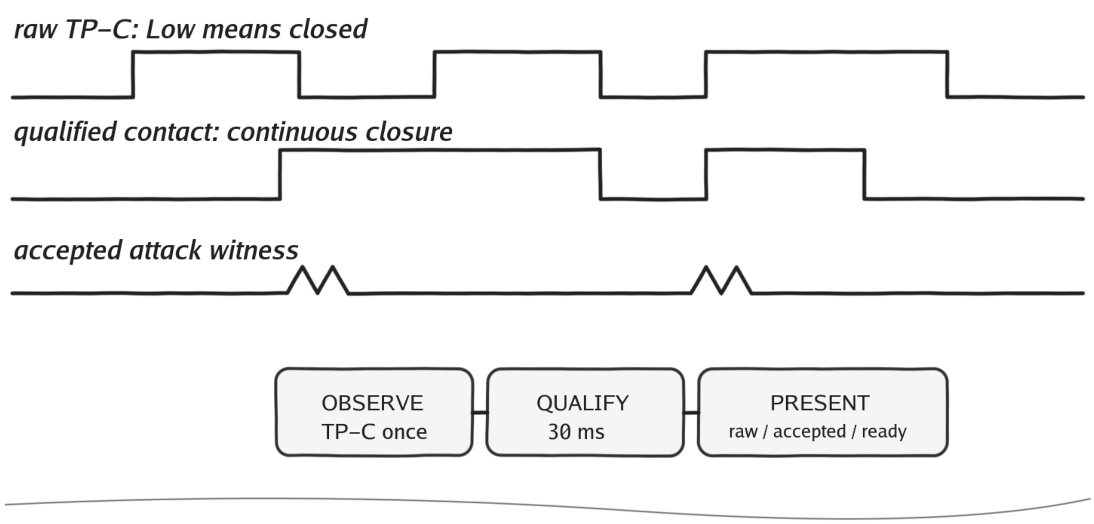
Reference gate. Use only a visibly identified C&K/Littelfuse `RB-220-07A R`, or a separately qualified substitution. Confirm the delivered part, terminal continuity, installed closure orientation, 10 kΩ resistor, and literal wiring before power. Remove USB power before wiring, moving the contact, or placing a probe. Stop on uncertain identity, damage, unexpected continuity, the wrong resistor, unresolved ground, or an out-of-bound prediction. The circuit and lesson are not physical acceptance.

1 Mission

raw level → qualified contact → attack event → release event.

`ContactDynamics` consumes copied samples. It does not measure force, acceleration, impact energy, orientation, damage, or safety. The canonical sketch keeps raw evidence visible while time policy rejects chatter and suppresses repeated attacks.

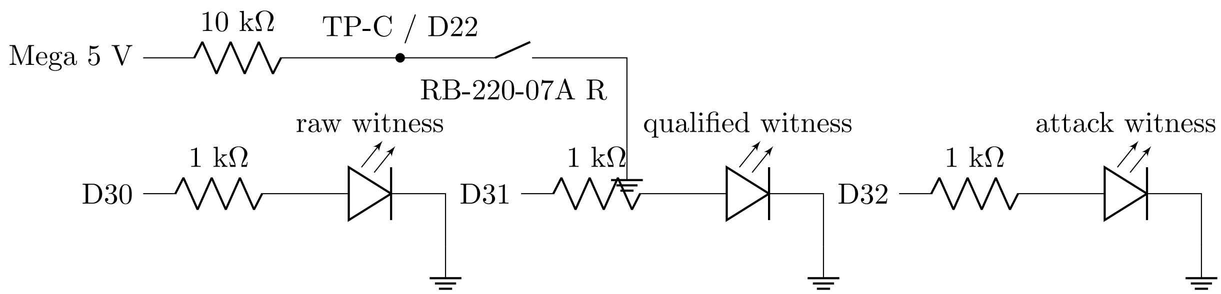
2 Inspect and orient



Alternative text: pencil timeline carries an active-low raw contact through continuous qualification, one accepted attack, refractory suppression, a qualified release, and a held-active warning.

The drawing is an orientation aid, not an electrical schematic. Match the exact part-number marking and reject damage, uncertain identity, mercury, unknown capsules, and bare piezo elements. The two switch terminals are electrically nonpolar, but installed orientation determines when the rolling ball closes them and must be recorded.

3 Electrically authoritative reference circuit



Formal schematic: the named manufacturer contact closes TP-C to ground; the external pull-up, not an internal pull, establishes High when open. Each evidence LED has its own current-limiting resistor.

At nominal 5 V, closure current is 5 V/10 kΩ = 0.5 mA. This lies between the datasheet's 10 μA minimum rated-load point and 1 mA maximum resistive load. E1 must measure the actual rail and resistor tolerance and prove worst-case current no greater than 1 mA.

Evidence	Mega pin	Meaning
TP-C	D22	High when open; Low when the reference contact closes
raw LED	D30 through 1 kΩ	sampled contact is closed
accepted LED	D31 through 1 kΩ	150 ms witness for an accepted attack
ready/fault LED	D32 through 1 kΩ	acquired and healthy; dark on stuck/fault

Configure `DigitalInput` with `Pull::None`; configure `ContactDynamics` active-low.

4 Policy contract

Construction is inert. `initialize()` validates nonzero, half-range-safe durations. `update(sample)` processes one copied level, timestamp, and source status. `snapshot()` is stable and non-consuming; `reset()` clears history and counters.

An attack requires continuous activity through `qualify`; release requires continuous inactivity through `release`. Refractory begins at an accepted attack. Completion before its exact boundary is suppressed, while completion at the boundary is accepted. A held contact creates no repeated attacks and reaches `StuckActive` at its configured boundary.

Source and timing faults create no event and require reset. Natural 32-bit rollover is valid. A half-range or larger apparent jump is invalid. An identical same-time sample is idempotent; changed same-time evidence faults.

5 Narrative Mega example

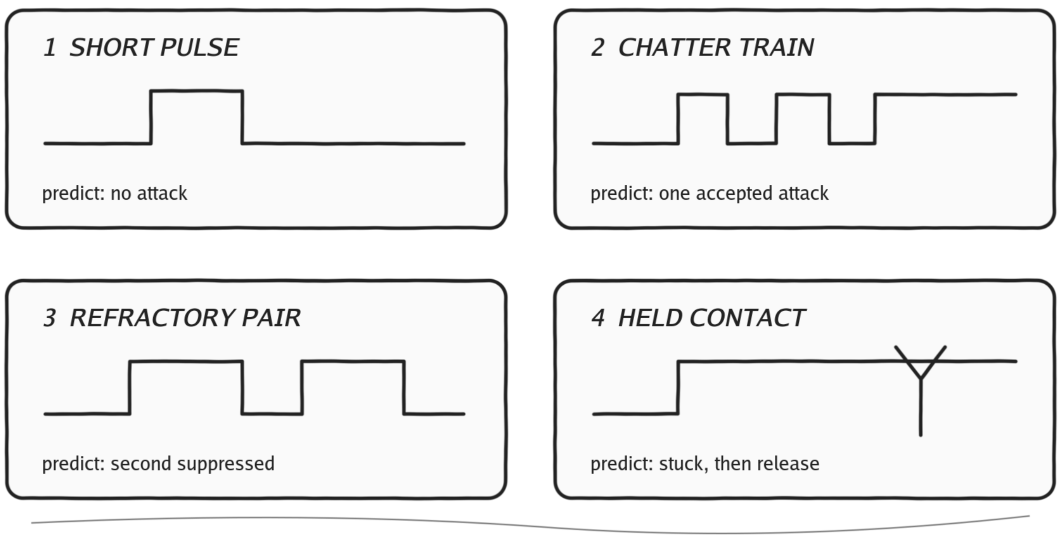
```
void loop ()
{
    const adk::TimePoint now (millis ());

    if (now.elapsedSince (startedAt).milliseconds () >=
```

```
            observationDurationMs
        || !observeContact (now) || !qualifyDynamics ()
        || !presentContactEvidence (now) || !actuateEvidence ())
    {
        stopSafely ();
    }
}
```

The sketch acquires D22 and all three LEDs, configures a bounded run, shows ready, then preserves observe–qualify–present–actuate order. Any failure and the 120-second boundary call `stopSafely()`. Serial is unnecessary.

6 Absent-hardware trace laboratory



Alternative text: four pencil laboratory cards show a short pulse rejected before qualification, chatter settling into one attack, a second attack suppressed during refractory, and a held contact warning.

No physical contact is needed. Use active-low policy with 30 ms qualification, 20 ms release, 80 ms refractory, and 120 ms stuck-active time.

Time (ms)	Level	Prediction
0	High	idle; counts 0/0
10	Low	active candidate starts
20	High	chatter cancels candidate
30	Low	candidate restarts
60	Low	accepted attack; counts 1/0
70	High	release candidate
90	High	release; pulse width 30 ms
100	Low	candidate during refractory
130	Low	suppressed attack; counts 1/1

Save complete status and snapshot fields after every row. Reset, replay, and compare fields rather than C++ object bytes. This establishes deterministic policy only; sampling cannot reveal a pulse entirely between polls.

Run a separate shortened E0 held-contact trace with `stuckActive=120 ms`: `(0,High)`, `(10,Low)`, `(40,Low)`, `(160,Low)`, `(170,High)`, and `(190,High)`. Predict acceptance at 40 ms, `StuckActive` exactly at 160 ms, and release with 150 ms pulse width at 190 ms. The physical sketch deliberately uses a longer 2 s boundary.

7 Predict, observe, interpret

1. Before power, predict open and closed TP-C levels, all three LEDs, and closed current. Inspect the part and resistor; unresolved identity stops.
2. Observe TP-C relative to Mega ground, then raw, accepted, and ready/fault LEDs. Record time and do not revise the prediction.
3. Interpret a discrepancy by separating circuit level, sampling, and policy. Do not infer physical intensity from an accepted attack.

Acquisition and safe state are separate evidence. Successful start proves that D22 and the three output claims were acquired. Bench work must independently measure that shutdown releases every pin to high impedance.

8 Diagnosis

Observation	Inspect next
TP-C never changes	part identity, continuity, orientation, ground path
raw LED disagrees with TP-C	D22 mapping, active level, sample timing
accepted LED never lights	continuous duration and chatter
several accepted pulses from one hold	refractory and held-contact policy
all witnesses go dark	acquisition, source status, bounded stop

9 Open bench acceptance

Record the exact delivered part, board revision, supply and instrument, contact-resistor measurement, each LED resistor and current bound, predicted levels/current, observed TP-C and LED states, worst-case closed current, and shutdown high impedance. Leave every row blank until a human performs it. Compilation, trace replay, and this PDF are not physical verification.

10 References

- C&K/Littelfuse, RB series manufacturer datasheet.
- `src/contact_dynamics.h` and `tests/test_contact_dynamics.cpp`.
- `examples/Lesson037ContactDynamics/`.

Result. Host replay and Mega compilation establish the software path for the documented reference design. Incoming conformance, powered characterization, current measurement, safe-state measurement, and bench acceptance remain open.